

■ B.5.2 Assignments

$lhs = rhs$	immediate assignment: rhs is evaluated at the time of assignment
$lhs := rhs$	delayed assignment: rhs is evaluated when the value of lhs is requested

The two basic types of assignment in *Mathematica*.

Assignments in *Mathematica* specify transformation rules for expressions. Every assignment that you make must be associated with a particular *Mathematica* symbol.

$f[args] = rhs$	assignment is associated with f
$t/: f[args] = rhs$	assignment is associated with t
$f[g[args]] \wedge= rhs$	assignment is associated with g

Assignments associated with different symbols.

In the case of an assignment like $f[args] = rhs$, *Mathematica* looks at f , then the head of f , then the head of that, and so on, until it finds a symbol with which to associate the assignment.

When you make an assignment like $lhs \wedge= rhs$, *Mathematica* will set up transformation rules associated with each distinct symbol that occurs either as an argument of lhs , or as the head of an argument of lhs .

The transformation rules associated with a particular symbol s are always stored in a definite order, and are tested in that order when they are used. Each time you make an assignment, the corresponding transformation rule is inserted at the end of the list of transformation rules associated with s , except in the following cases:

- The left-hand side of the transformation rule is identical to a transformation rule that has already been stored, and any $/;$ conditions on the right-hand side are also identical. In this case, the new transformation rule is inserted in place of the old one.

- Transformation rules that are already in the list have left-hand sides which are patterns that match the left-hand side of the transformation rule being inserted. In this case, the new transformation rule is inserted immediately before the first transformation rule that matches it. (Neither new nor old transformation rules can have `/;` conditions attached in this case.)